# Reasoning Machines

## Distillation of LLM into LRM

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**Abstract**

We distill large language models into large reasoning models via implicit recurrent deep learning. Attention and logic are entwined; the distributions of Transformer attention query updates over keys define a compositional relation between judgements in Open Horn Type Theory (OHTT). Dependent type theory OHTT is sound, has semiotic logical hexagon reasoning, and enables abductive inference that reverses entailment.

## Introduction

Attention and logic are intimately entwined; the distributions of Transformer attention query updates over keys define a compositional relation between judgements in Open Horn Type Theory (OHTT). OHTT is sound, provable formulas are valid - their semantic interpretations are true in all models. Moreover, this dependent type theory has semiotic logical hexagon reasoning, and most importantly, enables abductive inference, the reverse of entailment. Based on this observation, we propose distilling large language models into large reasoning models via implicit recurrent deep learning.

## Notation

[n] := {1, 2, ... , n}
α_i, j denote 'alpha subscript i, j'
α^h; k denote 'alpha superscript h, subscript k'
Σ iε[n] denote 'the sum over i, from 1 to n'

## LLM Attention

To set the stage, we adopt Thickstun's formalization [1] of Transformer networks, so as to be generic across LLM implementations.

A transformer block is a parameterized function class

fθ : R^pxd→ R^p×d. If x ∈ R^p×d then fθ(x) = z where:

dx1 token x_i
kx1 queries Q^h(x_i) = W^T; h,q x_i,
kx1 keys K^h (x_i) = W^T; h,k x_i,
kx1 values V^h(x_i) = W^T; h,v x_i,
weight matrices W_h,q, W_h,k, W_h,v ∈ R^d×k, (1)

probabilities α^h; i,j =
softmax _j ( < Q^h(x_i) , K^h (x_j) > /√k ), (2)

u′_i = Σ hε[H] W^T; c,h Σ jε[p] α^h; i,j V^h(x_j)
where W^T; c,h ε R^kxd, (3)

ui = LayerNorm(x_i + u′_i ; γ_1, β_1)
where γ_1, β_1 ∈ R, (4)

z′_i = W^T; 2 ReLU(W^T; 1 u_i)

where $W_1 \in R^{d\times m}$, $W_2 \in R^{m\times d}$, (5)

dx1 contextual representation of $x_i$
$z_i = \text{LayerNorm}(u_i + z'_i; \gamma_2, \beta_2)$
where $\gamma_2, \beta_2 \in R$. (6)

The LayerNorm function is defined for $z \in R^k$ by

$\text{LayerNorm}(z; \gamma,\beta) = \gamma(z-\mu z)/\sigma z + \beta$, $\gamma,\beta \in R^k$. (7)

$\mu z = 1/k \Sigma i\varepsilon[k] z_i$, $\sigma z = \text{sqrt}\ (\ 1/k \Sigma i\varepsilon[k] (z_i - \mu z)^2\ )$. (8)

The parameters θ consist of the entries of the weight matrices W, along with the LayerNorm parameters γ and β indicated above. The input $x \in R^{p\times d}$ should be interpreted as a collection of p objects, each with d features (often, but not always, a length-p sequence of d-vectors). Observe that the output $z \in R^{p\times d}$ has the same structure as the input $x \in R^{p\times d}$; a transformer is a composition of L transformer blocks, each with their own parameters: $f\theta_L \circ\cdots\circ f\theta_1 (x) \in R^{p\times d}$.
The hyper-parameters of the transformer are d, k, m, H, and L.

For notational simplicity, we now set H =1 and suppress the head indices. An input token $x_i$ gets updated to $z_i$, after L attention layers. Our distribution of interest, that links attention and logic, will be α the distribution of queries over keys, but not in its present form, we need an implicit recurrent deep learning proxy to carry out entailment and abduction, a good segue to our logic.

**Open Horn Type Theory (OHTT)**
Poernomo, 2025 [2]
Let J range over judgments. J may be any judgment form in any formal system. In a type-theoretic setting, these include formation judgments (A type), typing judgments (t : A), and equality judgments($s \equiv t : A$). Let Γ be a context — a finite, consistent collection of assumptions and prior witnesses.

There are two primitive judgment forms:
$\Gamma \vdash_+ J$ and $\Gamma \vdash_- J$
The first is read: “J is witnessed coherent in context Γ.” The second: “J is witnessed gapped in

context Γ.″

Both forms are proof-relevant. If γ : Γ ⊢+ J and γ′ : Γ ⊢+ J, we do not assume γ = γ′. Different witnesses to the same coherence may carry different structure. The same holds for gap witnesses: ω and ω′ witnessing the same gap may differ.

A judgment J is said to be open in context Γ if Γ contains neither a coherence witness for J nor a gap witness for J.

Exclusion Axiom: Coherence and gap, for the same judgment in the same context, are mutually exclusive.

Let ⇒ denote a compositional relation between judgments. A semantic instantiation of OHTT is entailment: J ⇒ K if the meaning of J implies the
meaning of K.
The specific choice of ⇒ depends on the semantic domain application, for example word-tokens.

For judgments J,K,L and compositional relation ⇒, the open horn type is:
Λ(J,K, L) := (Γ ⊢+ J ⇒ K)×(Γ ⊢+ K ⇒ L)×(Γ ⊢− J ⇒ L)
An inhabitant of Λ(J,K,L) is a triple (γ1,γ2,ω) where γ1 witnesses that J composes to K, γ2 witnesses that K composes to L, and ω witnesses that the direct composition from J to L is gapped.

A formal semantic interpretation of OHTT [2]:
Definition (Interpretation).
An OHTT context Γ is interpreted as a ruptured Kan complex Γ .
A type A in context Γ is interpreted as a ruptured fibration pA : Γ.A→Γ .
A term t : A in context Γ is interpreted as a coherent section: a map st : Γ→Γ.A with pA∘st =id and st landing in Coh.

Theorem (Soundness).
If Γ ⊢+ J is derivable in OHTT, then J holds in the interpretation.
If Γ ⊢− J is derivable in OHTT, then J is gapped in the interpretation.
Coherence and gap cannot both be derivable (Exclusion).
Proof sketch. Induction on derivations. Each OHTT rule corresponds to a construction on ruptured fibrations that preserves coherence or gap as appropriate. The Exclusion Law

corresponds to the exclusion condition in ruptured simplicial sets.

Let A be a space (Kan complex) modeling a semantic domain—say, the space of word-tokens. We now introduce Eve a semiotic reasoning model on top of a LLM but first:

**Trichotomy mnemonic***

Coherent: P *entails* Q iff ‘Q is a subset of P’

Gapped: P *contradicts* Q iff ‘P Λ Q and Q - P are non-empty’

Open: P and Q are *neutral* iff ‘P Λ Q is empty’

*We will explain the ‘ … ‘ shorthand in the following Eve LLM semantics.

**LLM Semantics**

We first define Eve, an implicitRNN [3] to proxy LLM contextual representations z,
as follows:
For i ε [m], tokens x_i ε R^p.
Initial condition: h_0 := 0 ε R^q
Input: u_i := [ x_i h_i-1 ]’
Equilibrium equation:
s_i = ϕ( A s_i + B u_i )
Prediction equation:
h_i = S(h_i-1, x_i) = C s_i + D u_i
(with parameterization [8] A := Γ- diag( |Γ| 1 )+ γ Id, where
0 < γ < 1, ensures A well-posed for nonlinear activation ϕ, that is, the fixed point equation has a solution and it is unique).

Eve training problem:
Let λ > 0, learn matrices A, B, C, D, and fixed points s_j, to minimize:

Σ i ε [m] (1 - Cos( h_i, z_i )) (*)

where Cos(a,b) = <a, b>/(||a|| ||b||),
to make h_i = Eve(x_i) a recurrent proxy of update z_i.

Entailment [4]:

Suppose Eve recursively processes two token string sentences,
premise P = (p_1, ... , p_r) and hypothesis H = (h_1, ... , h_s) as follows:
h_P,j = S(h_P,j-1, p_j), j ε [r] and
h_Q ,j = S(h_Q,j-1, q_j), j ε [s],
to obtain h_P := h_P,r and h_Q := h_Q,s.

Define probabilities β(h) = softmax _j ( < Q(h), K(z_j) > /√k ),
β(h) = (β_1(h), ... , β_m(h)) is the distribution
of an updated query over keys k_j = K(z_j).
Fix 0 < τ < 1. Define F_τ(h) := { k_j | β_j(h) > τ }.
Using a decision tree, we now explain the trichotomy mnemonic shorthand.

Define **entailment** denoted P =>_τ Q iff
F_τ(h_Q) is a subset of F_τ(h_P).
We will drop the subscript τ unless necessary.

Define **contradiction** denoted P => not Q iff
F_τ(h_Q) is a not subset of F_τ(h_P) and
F(h_P) Λ F(h_Q) is non-empty.

P and Q are said to be **neutral** iff
intersection F(h_P) Λ F(h_Q) is empty.

On training triples (P, Q, y), for example from the
Stanford Natural Language Inference (SNLI) Corpus,
where y ε { entail, neutral, contradiction}
learn **Entailment** defined by:

Entailment(h_P, h_Q) :=
entail, if P => Q,
contradiction, if P => not Q,
neutral otherwise,

to minimize (*) and Entailment loss,
(for example, # entails misclassified as a contradiction +
# contradictions misclassified as an entail +
# neutrals misclassified as entail or contradiction).
Thereby, Eve learns textual entailment.

Theory OHTT has a simplicial model semantics, Kan complexes of word-tokens are a domain application. The OHTT compositional relation ⇒ can then be interpreted as entailment in Eve, enabling logical reasoning.

Suppose in Eve, P =>_τ Q and Q =>_μ R, what is the semantics of open horn type
Λ(P, Q, R) := (Γ ⊢ + P ⇒ Q)×(Γ ⊢ + Q ⇒ R)×(Γ ⊢ – P ⇒ R)?

i) if μ = τ then P =>_τ R, the horn is coherently filled,
ii) if μ > τ then writing < for 'is a subset of'
F_μ(h_R) < F_τ(h_R) and
F_μ(h_R) < F_μ(h_Q) < F_τ(h_Q) < F_τ(h_P) hence F_μ(h_R) < F_τ(h_P) Λ F_τ(h_R),
the horn is gap witnessed,
iii) otherwise μ < τ, neither coherence nor gap is witnessed for P ⇒ R, and the horn is open.

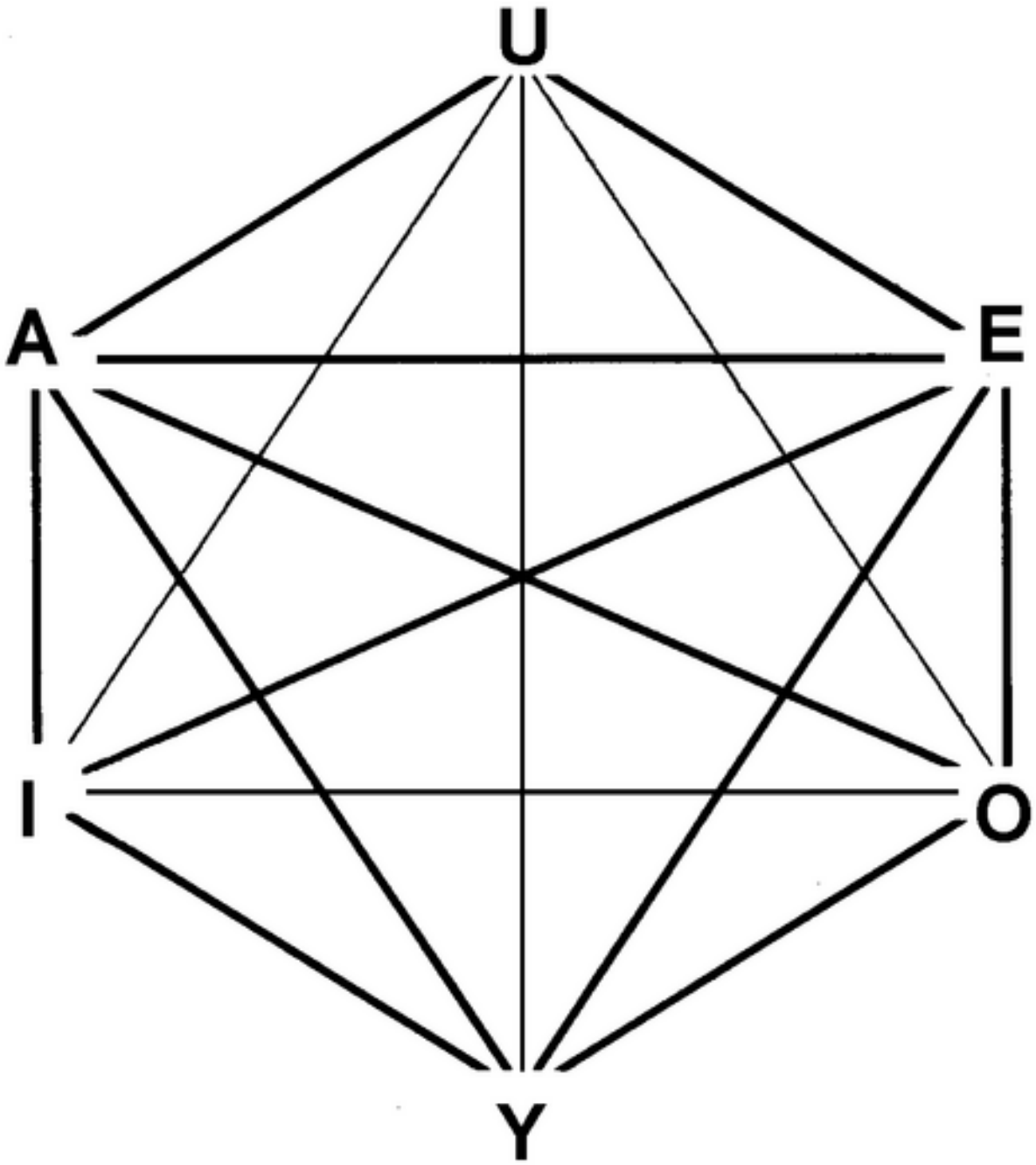
Curiously, attention and logic are entwined by β, a function of the similarity function chosen, here scaled dot product. (Fibrations are both ubiquitous and foundational.)

Abductive inference [4]:
Given a hypothesis Q, we wish to find a premise P such that P => Q. Token string Q may be auto-regressively continued by Eve to generate Q, P1, P2, ... , Pn, where n > 1, and Pj denotes a continuation sentence. Choose 0 < τ < 1, for each j ε [n],
check if Pj => Q. Suppose Ω = { Pj | Pj => Q } is non-empty.
Let m := min over P ε Ω, | F(h_P) | and Pj* be the first Pj In Ω with | F(h_Pj) | = m.

Abduction: Given a hypothesis Q, we want to determine a premise P that entails Q. If Ω is non-empty, choose premise Pj* as above that entails Q. Pj* => Q is a parsimonious selection of a premise to explain Q.

So Eve has textual entailment and abductive reasoning.

**Semiotic Logical Hexagon** [5] for OHTT




A := Coherence witness $\gamma : \Gamma \vdash + P \Rightarrow Q$

E := Gap witness $\omega : \Gamma \vdash - P \Rightarrow Q$

$I = \neg E$ and $O = \neg A$

$U = A \vee E$ - Determinacy, either coherent or gapped or both

$Y = I \wedge O$ - Open, neither coherence nor gap is witnessed for P ⇒ Q

Semiotics [6]
Suppose random variable X has distribution p_X,
denoted X ~ p_X.
Let H_α denote Renyi α-entropy.
For (X, Y) ~ p_X,Y = p_X p_Y|X, Arimoto mutual information
I_A,α (X; Y) := H_α(X) - H_A,α(X|Y)
where H_A,α is Arimoto conditional α-entropy.

To quantify the uncertainty of text, define semiotic breadth
S(Q) := H_α(h_Q) where h_Q ~ β(h_Q).
We say P is an 'interpretation' [2] of Q if P => Q
and define semiotic decipherability D(P) := I_A, α(h_Q; h_P)
to quantify the reduction in uncertainty about an interpretation
that results from observing a message.
To profile semiotic channel capacity C_A,α := max over h_Q, I_A, α(h_Q; h_P)
map S against D, creating a LLM semiotic cloud signature.

**Conclusion**
Implicit recurrent deep learning can distill a large language model into a reasoning model, by proxying attention mechanism contextual representations of tokens. The support of a distribution of a proxy over keys induces semantic entailment of composable OHTT judgements, enabling semiotic logical hexagon reasoning, and importantly, abductive inference that reverses entailment.

Note that we have the flexibility of Eve's semantic reasoning in natural language with learned textual entailment, or more constrained formal syntactic reasoning in open horn type theory, or both - surmise then verify. Abductive inference has application to **reasoning copilots** capable of tasks such as suggesting medical differential diagnoses.

In agentic systems, agents communicate and coordinate their actions through large language models with workflow protocol guardrails. The design of testbeds is difficult because of the unpredictability of emergent swarm behavior. An application of Eve large reasoning models is white hat hacking, as they are inherently auditable; they show and explain their work, through their state trajectories. Recall state h_i is a proxy contextual representation of token x_i. Agent

**states can combine to produce offspring via state combination operators [4] like convolution, Kronecker product, Hadamard product, and rank-one operators, enabling genetic algorithm selection and system evolution, opening the possibility of swarm intelligence.**

**In summary, LRM Eve on top of a LLM, is able to reason via Open Horn Type Theory and infer abductively via textual entailment; a computational epistemology independent of mind or intent. Minds express meaning in models of theories. Sound interpretations enable syntactic reasoning about model semantics. Entailment enables abductive reasoning and the semiotic logical hexagon enables counterfactual reasoning, all by machine. Eve pointing to the moon. Sometimes we need a quick LLM answer, sometimes we need a careful LRM reason, to wit Kahneman's thinking, fast and slow! [7]**

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